

Vani Agarwal

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EDUCATION

University of Washington - Seattle

Mathematics (B.A.) & Informatics (B.S.)

Relevant Coursework: Data Structures & Algorithms, Database Management, Software Design & Implementation, Data Visualization, Operating Systems, Web Development

Graduation: June 2026 - Eligible for 2026 start

SKILLS

Languages: Java, Python, C#, SQL, JavaScript, HTML/CSS, R

Frameworks/Libraries: React, Node.js, Flask, Supabase, Firebase, Azure, Git, GitLab CI/CD, Postman, IntelliJ, Visual Studio

Certifications: Google Data Analytics Professional Certificate, The Complete Web Developer ZTM

Concepts: Scalable API Development, Distributed Systems, Data Modeling, Secure Authentication, UI/UX Design, Algorithm Optimization

EXPERIENCE

Backend Engineering Intern

Jun. 2025 - Aug. 2025

United Postal Services

Atlanta, GA

- Designed and implemented **two REST APIs** enabling secure retrieval of **10K+ document metadata records** and **50K+ PDF payloads** through a unified interface, reducing retrieval time by 40%.
- Integrated JWT-Bearer authentication with Azure AD, strengthening access control and security for backend services.
- Collaborated with cross-functional teams to improve API design, contributing to scalable enterprise solutions.

Software Engineering Intern

Jan. 2025 - Mar. 2025

Rex

Remote

- Built a scalable social recommendation platform, enabling users to share and discover products via personalized feeds and groups.
- Developed a mobile-first **React frontend** and integrated **secure user authentication** using **Supabase Auth**.
- Architected a normalized **PostgreSQL schema** for user profiles, groups, & preferences to support future expansion.

Software Engineering Intern

Aug. 2024 - Dec. 2024

Societal

Seattle, WA

- Developed an automated OCR pipeline processing **5K+ financial documents/month**, with custom table detection algorithms for accurate data extraction.
- Designed validation pipelines that improved data quality, reducing post-processing errors by **35%**.
- Experimented with multiple OCR techniques, boosting extraction accuracy from **82% to 94%**.

Computer Science & Engineering Teaching Assistant

Mar. 2024 - Present

University of Washington | Data Structures & Algorithms, Intermediate Data Programming

Seattle, WA

- Analyzed **5K+ student performance data points** to identify learning gaps, informing course content updates.
- Authored **JUnit test suites** to evaluate algorithm correctness and efficiency for **300+ student submissions/quarter**.
- Leveraged GitLab CI/CD for automated testing, identifying bottlenecks, offering optimization advice.

PROJECTS

Remedii | React, Firebase, Firestore, OpenAI GPT-4, OAuth

- Developed a **responsive med-tracking app**, featuring customizable medication reminders & interaction alerts.
- Utilized **Firebase** for secure **OAuth authentication** and **Firestore** for **real-time database management**.
- Leveraged **GPT-4** for **natural language processing** to deliver personalized health insights.
- Designed and implemented a **React frontend**, focused on **intuitive UI/UX**, improving accessibility & usability.

Education Inequality in the US | Javascript, R, Vega-Lite

- Designed an interactive data visualization platform using **ObservableHQ**, **D3.js**, and **JavaScript**, analyzing educational spending and literacy trends across **42 U.S. states**.
- Optimized** data processing and visualisation, reducing latency by **20%** for datasets exceeding **500K entries**.
- Implemented **dynamic user interactions** to enhance **data exploration**, improving engagement by **30%**.

Geospatial Data Analysis & Visualization | Python, GeoPandas, Matplotlib, Pandas, Folium

- Developed a data analysis pipeline using **Python**, **GeoPandas**, **Pandas**, and **Folium** to study food access in Washington State, integrating **2010 U.S. Census** and **USDA ERS data**.
- Merged geospatial shapefiles and tabular datasets, analyzing **1,318 census tracts**, & generated **choropleth maps** and population-based visualizations with **Matplotlib**, identifying regions with low access to food based on thresholds.